Akoizo Original Practice - Form 1 - Answer key

Division B | Crime Busters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which solid turns brown iodine colorless?

Unit: Identifying white powders.
Worked-case data: An unknown white powder is insoluble in water, fizzes strongly with HCl, and leaves iodine yellow-brown. Identify it.
Separate investigation observations:
Appearance: Fine, pure white powder that squeaks when pressed.
Water: Does not dissolve; forms a cloudy suspension.
HCl: No fizzing.
Iodine: Turns blue-black.

A. Vitamin C

B. Cornstarch

C. Table salt

D. Gypsum

Correct: A

2. (2 points) A human hair usually has a medullary index of:

Unit: Hair and fibers.
Worked-case data: A hair has a continuous medulla about 60% of the hair’s width and scales shaped like petals. What is its source?
Separate investigation observations:
Hair A: Medulla fragmented; index about 0.2.
Hair B: Wide, continuous ladder-like medulla; flattened scales.
Fiber: Burns quickly with afterglow and soft gray ash.

A. Zero always

B. Less than about one-third

C. Greater than one-half

D. Exactly one

Correct: B

3. (2 points) Why is the starting line drawn in pencil?

Unit: Paper chromatography.
Worked-case data: Ransom-note ink separates into a blue spot at 2.4 cm and a yellow spot at 4.8 cm, with the solvent front at 6.0 cm. Suspect A's pen gives Rf values 0.40 and 0.80; suspect B's gives 0.40 and 0.65. Which pen matches?
Separate investigation observations:
Note ink: Three bands: Rf 0.15, 0.45, 0.70.
Suspect 1 pen: Two bands: Rf 0.45, 0.70.
Suspect 2 pen: Three bands: Rf 0.15, 0.45, 0.70.
Suspect 3 pen: Three bands: Rf 0.20, 0.50, 0.90.

A. Ink is required for Rf

B. Pencil changes Rf values

C. Pencil graphite does not dissolve and separate

D. Pencil speeds up the solvent

Correct: C

4. (2 points) An analysis must explain the culprit and also:

Unit: Writing the crime scene analysis.
Worked-case data: Evidence: the scene powder is baking soda; suspect A has baking soda, B has cornstarch, C has baking soda. Note ink matches only C's pen. A dog hair was found; A and C own dogs, B owns a cat. Who is the culprit?
Separate investigation observations:
Powder: Calcium carbonate: found with suspects 1 and 3.
Plastic: HDPE fragment: suspects 1, 2, and 3 all use HDPE jugs.
Chromatography: Note ink matches only suspect 3's pen.
Fingerprint: Scene loop matches suspect 3's minutiae.

A. Only the strongest evidence

B. Each suspect's alibi only

C. The price of each test

D. Why the other suspects were not chosen

Correct: D

5. (2 points) Which liquid is basic with a strong, pungent odor?

Unit: Metals and liquids.
Worked-case data: A silvery metal is not magnetic, reacts vigorously with HCl, and has a density of about 1.7 g/cm^3. Identify it.
Separate investigation observations:
Liquid color: Clear and colorless.
Odor: Little odor.
pH paper: About pH 6, close to neutral.
With yeast: Vigorous foaming and bubbling.

A. Hydrogen peroxide

B. Lemon juice

C. Household ammonia

D. Vinegar

Correct: C

6. (2 points) Which plastic gives a green flame in the Beilstein test?

Unit: Recyclable plastics.
Worked-case data: A plastic fragment sinks in water, floats in a 1.10 g/cm^3 salt solution, and burns with a sooty flame. Identify it.
Separate investigation observations:
Water: Sinks.
1.10 g/cm^3 solution: Sinks.
1.25 g/cm^3 solution: Floats.
Appearance: Very clear, rigid, and shatter-resistant.

A. PETE

B. HDPE

C. PP

D. PVC

Correct: D

7. (2 points) An elongated stain’s narrow tail points:

Unit: Fingerprints, DNA, prints, soil, and spatter.
Worked-case data: A scene DNA sample shows peaks at 12 and 15 repeats at location 1, and 8 and 8 at location 2. Suspect X: 12/15 and 8/10. Suspect Y: 12/15 and 8/8. Which suspect matches?
Separate investigation observations:
Scene print: A whorl with two deltas.
Suspect A: Right index finger: loop.
Suspect B: Right index finger: whorl with matching ridge endings.
Suspect C: Right index finger: arch.

A. In the direction of travel

B. Back toward the source

C. Straight up

D. In a random direction

Correct: A

8. (2 points) Which two solids give a blue-black iodine result?

Unit: Identifying white powders.
Worked-case data: An unknown white powder is insoluble in water, fizzes strongly with HCl, and leaves iodine yellow-brown. Identify it.
Separate investigation observations:
Appearance: Fine, pure white powder that squeaks when pressed.
Water: Does not dissolve; forms a cloudy suspension.
HCl: No fizzing.
Iodine: Turns blue-black.

A. Salt and sugar

B. Sand and gypsum

C. Yeast and gelatin

D. Flour and cornstarch

Correct: D

9. (2 points) A fiber melts, shrinks from the flame, and forms a hard bead. It is:

A. Synthetic

B. Vegetable

C. Animal

D. Hair

Correct: A

10. (2 points) A component that clings strongly to the paper will have:

Unit: Paper chromatography.
Worked-case data: Ransom-note ink separates into a blue spot at 2.4 cm and a yellow spot at 4.8 cm, with the solvent front at 6.0 cm. Suspect A's pen gives Rf values 0.40 and 0.80; suspect B's gives 0.40 and 0.65. Which pen matches?
Separate investigation observations:
Note ink: Three bands: Rf 0.15, 0.45, 0.70.
Suspect 1 pen: Two bands: Rf 0.45, 0.70.
Suspect 2 pen: Three bands: Rf 0.15, 0.45, 0.70.
Suspect 3 pen: Three bands: Rf 0.20, 0.50, 0.90.

A. No spot at all

B. A low Rf

C. A high Rf

D. An Rf above 1

Correct: B

11. (2 points) Which phrase is best for a fiber match?

Unit: Writing the crime scene analysis.
Worked-case data: Evidence: the scene powder is baking soda; suspect A has baking soda, B has cornstarch, C has baking soda. Note ink matches only C's pen. A dog hair was found; A and C own dogs, B owns a cat. Who is the culprit?
Separate investigation observations:
Powder: Calcium carbonate: found with suspects 1 and 3.
Plastic: HDPE fragment: suspects 1, 2, and 3 all use HDPE jugs.
Chromatography: Note ink matches only suspect 3's pen.
Fingerprint: Scene loop matches suspect 3's minutiae.

A. “Identical source”

B. “Certainly from”

C. “Consistent with”

D. “Proves beyond doubt”

Correct: C

12. (4 points) Why might aluminum react slowly with HCl at first?

Unit: Metals and liquids.
Worked-case data: A silvery metal is not magnetic, reacts vigorously with HCl, and has a density of about 1.7 g/cm^3. Identify it.
Separate investigation observations:
Liquid color: Clear and colorless.
Odor: Little odor.
pH paper: About pH 6, close to neutral.
With yeast: Vigorous foaming and bubbling.

answer: A thin aluminum oxide layer protects the metal until the acid dissolves it, after which the reaction speeds up. (4 points).

13. (4 points) Why must air bubbles be removed before a density test?

Unit: Recyclable plastics.
Worked-case data: A plastic fragment sinks in water, floats in a 1.10 g/cm^3 salt solution, and burns with a sooty flame. Identify it.
Separate investigation observations:
Water: Sinks.
1.10 g/cm^3 solution: Sinks.
1.25 g/cm^3 solution: Floats.
Appearance: Very clear, rigid, and shatter-resistant.

answer: Trapped air lowers the overall density of the sample, which can make a dense plastic float falsely. (4 points).

14. (4 points) Explain the difference between class and individual evidence, with an example of each.

Unit: Fingerprints, DNA, prints, soil, and spatter.
Worked-case data: A scene DNA sample shows peaks at 12 and 15 repeats at location 1, and 8 and 8 at location 2. Suspect X: 12/15 and 8/10. Suspect Y: 12/15 and 8/8. Which suspect matches?
Separate investigation observations:
Scene print: A whorl with two deltas.
Suspect A: Right index finger: loop.
Suspect B: Right index finger: whorl with matching ridge endings.
Suspect C: Right index finger: arch.

answer: Class evidence narrows to a group, like a shoe brand and size. Individual evidence can point to one source, like a full fingerprint or a complete DNA profile. (4 points).

15. (4 points) Explain why sodium acetate gives a vinegar smell with HCl.

Unit: Identifying white powders.
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Separate investigation observations:
Appearance: Fine, pure white powder that squeaks when pressed.
Water: Does not dissolve; forms a cloudy suspension.
HCl: No fizzing.
Iodine: Turns blue-black.

answer: HCl donates a proton to acetate, forming acetic acid, which is the acid in vinegar and has its characteristic odor. (4 points).

16. (4 points) How does wool look under a microscope compared with cotton?

Unit: Hair and fibers.
Worked-case data: A hair has a continuous medulla about 60% of the hair’s width and scales shaped like petals. What is its source?
Separate investigation observations:
Hair A: Medulla fragmented; index about 0.2.
Hair B: Wide, continuous ladder-like medulla; flattened scales.
Fiber: Burns quickly with afterglow and soft gray ash.

answer: Wool shows overlapping scales like hair because it is animal hair; cotton appears as a flat, twisted ribbon. (4 points).

17. (4 points) What makes two inks a match on a chromatogram?

Unit: Paper chromatography.
Worked-case data: Ransom-note ink separates into a blue spot at 2.4 cm and a yellow spot at 4.8 cm, with the solvent front at 6.0 cm. Suspect A's pen gives Rf values 0.40 and 0.80; suspect B's gives 0.40 and 0.65. Which pen matches?
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Suspect 2 pen: Three bands: Rf 0.15, 0.45, 0.70.
Suspect 3 pen: Three bands: Rf 0.20, 0.50, 0.90.

answer: The same number of bands, colors, order, and Rf values. (4 points).

18. (4 points) How should you handle a result that does not fit your conclusion?

Unit: Writing the crime scene analysis.
Worked-case data: Evidence: the scene powder is baking soda; suspect A has baking soda, B has cornstarch, C has baking soda. Note ink matches only C's pen. A dog hair was found; A and C own dogs, B owns a cat. Who is the culprit?
Separate investigation observations:
Powder: Calcium carbonate: found with suspects 1 and 3.
Plastic: HDPE fragment: suspects 1, 2, and 3 all use HDPE jugs.
Chromatography: Note ink matches only suspect 3's pen.
Fingerprint: Scene loop matches suspect 3's minutiae.

answer: Acknowledge it, look for an explanation, and state why it carries less weight than the stronger evidence. (4 points).